Week 10 Notes

STAT 251 SECTION 01

From estimating to testing

For the last two weeks we have been **estimating** parameters: a confidence interval answers the question *"what is p (or μ), give or take?"* A lot of questions are not asked that way. They come as a **claim**, and we want to know whether the data back it up:

- A new drug claims to lower blood pressure. Does it, or did the patients' readings just happen to drop?
- The polling averages say a Senate candidate has 52% support. A new poll finds less. Has support really dropped, or is the new poll just a different sample?
- Your roommate says they can tell Coke from Pepsi. Can they, or are they guessing?

Each claim is about a **parameter**: the drug's average effect μ, the candidate's support p, your roommate's chance p of naming a cup correctly. We never see the parameter. We see an **estimate** from a sample, and estimates bounce around from sample to sample (Week 8: two polls of the same state give two different answers). So the real question is always:

> **Is the difference between what we saw and what the claim says too big to be ordinary sampling variation?**

A **significance test** (also called a **hypothesis test**) is the tool that answers it. A result that is decidedly **not** due to ordinary variation, that is, not due to chance or coincidence, is called **statistically significant**.

The logic of a significance test

In plain words: a courtroom

A significance test works like a criminal trial.

- The defendant is **presumed innocent**. Nobody has to prove innocence; innocence is the starting assumption. In a test, the starting assumption is the boring explanation, "nothing

interesting is going on", and it is called the **null hypothesis**.

- The prosecution has a **suspicion**: the defendant did it. In a test, the suspicion is called the **alternative hypothesis**. The burden of proof is on the suspicion, not on the boring explanation.
- The jury asks: **if the defendant really were innocent, how likely is evidence like this?** If that is very unlikely (beyond reasonable doubt), they convict. In a test, that probability is the **P-value**, and "beyond reasonable doubt" is a cutoff we pick in advance, the **significance level** α.
- A jury that does not convict says **"not guilty"**, never "innocent". Weak evidence does not prove innocence. In a test, we either **reject** the null hypothesis or **do not reject** it. We never "accept" it.

Hold on to that picture. Everything below is the same four ideas with numbers attached.

Working it through: can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. You set up a blind test: 8 times, you flip a coin to decide which soda to pour, and they taste the cup and name it. They get **all 8 right**. Impressive, or lucky?

Step 1: name the parameter, and the two stories

The parameter is p, your roommate's true, long-run chance of naming a cup correctly. We cannot see p; we only see this one test, where they got $\hat{p} = 8/8 = 1$. There are two stories about p:

- **Story 1, "just guessing":** they cannot really tell, so each cup is a coin flip, and $p = 0.5$. This is the **null hypothesis**, written $H_0 : p = 0.5$.
- **Story 2, "they can taste it":** they do better than guessing, $p > 0.5$. This is the **alternative hypothesis**, written $H_A : p > 0.5$.

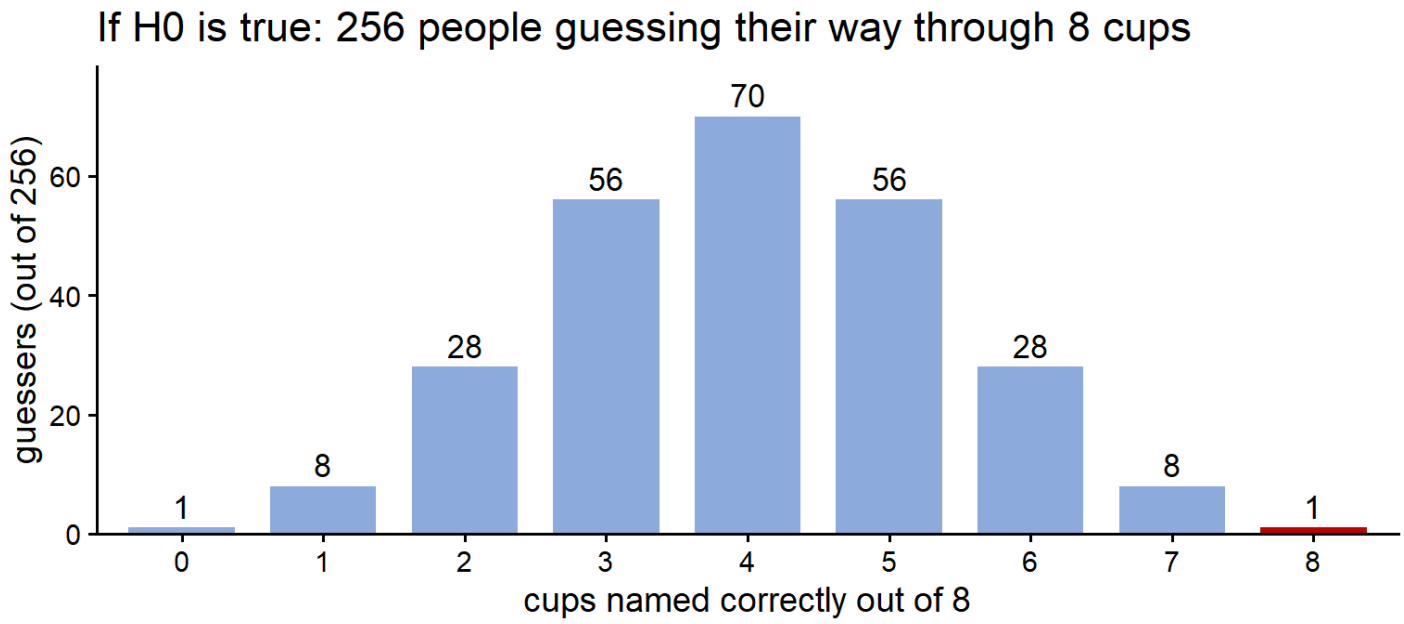
Notice that the hypotheses are about p, the parameter, not about $\hat{p}$. We already **know** $\hat{p} = 1$; there is nothing to test about it. The question is what it tells us about p.

Step 2: suppose H_0 is true. What results would we expect?

Here is the key move. We cannot prove story 2 directly, so we **assume story 1** and work out what it predicts. If they are only guessing, then the number of cups they get right, X, is a count of successes in 8 independent trials that each succeed with probability 0.5. That is a binomial random variable from Week 7:

$$X \sim \text{Binom}(8,\ 0.5) \qquad \text{when } H_0 \text{ is true}$$

A helpful way to see this distribution: imagine **256 people who cannot taste the difference** each take the 8-cup test, guessing every time. Since $2^8 = 256$, the binomial formula $P(X = k) = \binom{8}{k}(0.5)^k(0.5)^{8-k} = \binom{8}{k}/256$ says that, on average, exactly $\binom{8}{k}$ of them score k:



This is what "just guessing" looks like. Most guessers land near 4 out of 8, which makes sense: the mean of a binomial is $np = 8(0.5) = 4$. Some get 5 or 6 by luck. Getting **all 8** happens to about **1 guesser in 256**.

Step 3: how surprising is what we actually saw?

Now compare the data with the prediction. The probability, **if H_0 is true**, of a result at least as extreme as the one we saw is called the **P-value**. "At least as extreme" means at least as far in the direction H_A points to, which here is "as many correct, or more":

$$\text{P-value} = P(X \geq 8 \mid H_0 \text{ true}) = \binom{8}{8}(0.5)^8 = \frac{1}{256} \approx 0.0039$$

The bar $\mid$ is read **"given"**: the probability is worked out in the world where H_0 is true. That is the same conditional probability idea from Week 6.

A P-value of 0.0039 says: **if your roommate were only guessing, a result this good would happen less than half a percent of the time.**

Step 4: two explanations, and a decision

There are only two ways to explain a result with a small P-value:

1. **Something unusual happened.** Your roommate was guessing and got very lucky: they are the 1 in 256.
2. **The assumption behind the calculation is wrong.** H_0 is false; they really can taste the difference.

Both are possible. The test does not tell us which one is true. It tells us how hard we would have to believe in luck to keep believing H_0. The smaller the P-value, the harder that gets, and at some point explanation 2 becomes the more reasonable one.

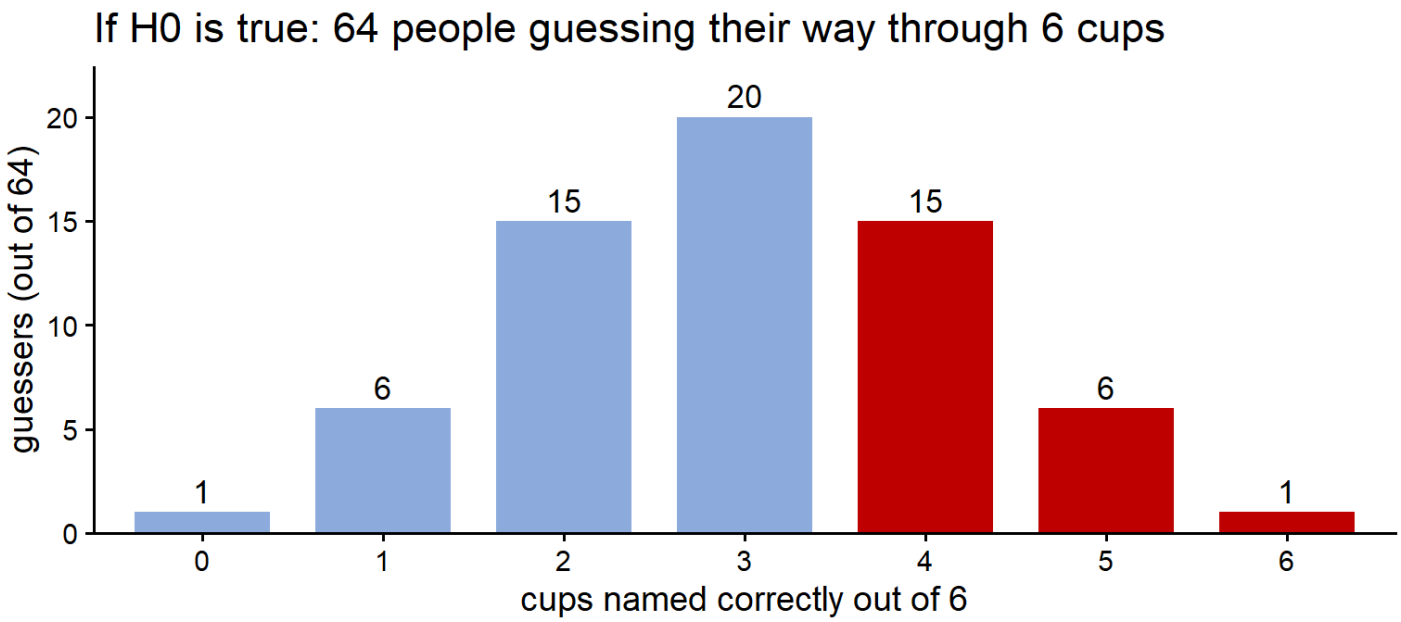
Where is that point? It is a choice, made **before** looking at the data, called the **significance level** α. By convention $\alpha = 0.05$, i.e. "if a result this extreme would happen less than 5% of the time by luck alone, we stop believing it was luck". The decision rule is:

- If the P-value $< \alpha$: **reject** H_0. The result is statistically significant; the data are evidence for H_A.
- If the P-value $\geq \alpha$: **do not reject** H_0. The result is not statistically significant; the data are not strong enough evidence for H_A.

Here $0.0039 < 0.05$, so we reject H_0: the data are evidence that your roommate can taste the difference. The textbook's memory device: **"if the P-value is low, the null must go."**

A second taster

A friend makes the same claim, but takes a shorter 6-cup test and gets **4 of 6** right. Under $H_0 : p = 0.5$, their score is $X \sim \text{Binom}(6,\ 0.5)$. Picture 64 guessers ($2^6 = 64$):



"At least as extreme" as 4 correct means 4, 5 or 6 correct, the shaded bars:

$$\text{P-value} = P(X \geq 4 \mid H_0 \text{ true}) = \frac{15+6+1}{64} = \frac{22}{64} \approx 0.34$$

About a third of pure guessers do this well, so 4 of 6 is not surprising at all if your friend is guessing. $0.34 \geq 0.05$: **do not reject** H_0. This does **not** prove your friend is guessing ("not guilty", not "innocent"). It says this test gives no real evidence that they can taste the difference. A longer test might.

One more thought. If all 256 guessers from the first picture took the 8-cup test, about one of them would score 8 for 8 and be convinced they have a gift. A small P-value means **either** rare luck **or** a false H_0, and from one test we cannot be sure which. That is why surprising results get repeated before anyone believes them.

The notation

Significance tests bring a lot of new symbols. Most of them are old ideas with new names.

Symbol	Read as	Meaning
θ	"theta"	a parameter, any parameter: stands for p or μ when we talk about tests in general
θ_0, e.g. p_0, μ_0	"theta-naught", "p-naught", "mu-naught"	the value of the parameter that H_0 claims, a number like 0.5 or 28 g

Symbol	Read as	Meaning
$\hat{\theta}$, e.g. $\hat{p}$, $\bar{x}$	"theta-hat", "p-hat", "x-bar"	the estimate of θ from our sample (Weeks 1 and 8)
$H_0 : \theta = \theta_0$	"H-naught: theta equals theta-naught"	the null hypothesis: "no effect", the parameter equals the claimed value
$H_A : \theta > \theta_0$	"H-A: theta is greater than theta-naught"	an upper-tailed alternative: we suspect the parameter is bigger
$H_A : \theta < \theta_0$	"H-A: theta is less than theta-naught"	a lower-tailed alternative: we suspect the parameter is smaller
$H_A : \theta \neq \theta_0$	"H-A: theta is not equal to theta-naught"	a two-sided alternative: we suspect it is different, in either direction
z_{obs}	"z observed"	the test statistic: how many standard errors $\hat{\theta}$ is from θ_0
$P(Z \geq z_{obs} \mid H_0 \text{ true})$	"the probability Z is at least z observed, given H-naught is true"	the P-value of an upper-tailed test
α	"alpha"	the significance level, the cutoff for the P-value, usually 0.05

The subscript 0 always marks "the value from the null hypothesis". In the taste test, θ is p, $\theta_0 = p_0 = 0.5$, and $\hat{\theta} = \hat{p}$.

Writing the hypotheses

Hypotheses come first, before any data are looked at. A good order to write them in:

1. **Start with H_A**: what do we suspect, or hope to find evidence for? "They can taste the difference" becomes $H_A : p > 0.5$.
2. **Then write H_0** as "that effect is not there", with an equals sign: $H_0 : p = 0.5$.

Rules that always hold:

- Both hypotheses are about a **parameter** (p or μ), never an estimate. $H_0 : \hat{p} = 0.5$ makes no sense: we already know $\hat{p}$.
- H_0 always contains the **equals sign**, and H_A never does.

- H_A must come from the question, **not from the data**. Looking at the data first and then pointing H_A the way the data went is a form of bias: it makes the evidence look stronger than it is.

The direction of H_A decides which kind of test it is:

Alternative	Name	Use it when the question says
$H_A : \theta > \theta_0$	upper-tailed (one-sided)	"more than", "better", "increased", "at least"
$H_A : \theta < \theta_0$	lower-tailed (one-sided)	"less than", "worse", "decreased", "fewer"
$H_A : \theta \neq \theta_0$	two-sided	"different", "changed", "is it fair?"

Example: stating hypotheses

a. *Your roommate names Coke or Pepsi. Do they do better than guessing?* p = their chance of naming a cup right. $H_A : p > 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**.

b. *A coin is spun on a table. Does it favor heads or tails?* p = the chance of heads. We do not suspect a direction, only "not fair": $H_A : p \neq 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **two-sided**.

c. *A chip bag says 28 g. Do the bags hold less than that on average?* μ = the mean weight of all bags. $H_A : \mu < 28$, $H_0 : \mu = 28$, with $\mu_0 = 28$ g: **lower-tailed**.

Measuring "how surprising": the test statistic

For the taste test we could list every possible score and add up the binomial probabilities. With hundreds or thousands of observations that gets impractical, but we already have a better tool: the **sampling distribution** and the **central limit theorem** from Week 8.

Recall what the CLT told us about $\hat{p}$: over all possible samples of size n, the values of $\hat{p}$ pile up in a roughly normal shape, centered at the true p, with standard error $SE = \sqrt{p(1-p)/n}$. In a significance test we **assume H_0 is true**, so the true p is p_0. That pins down the whole pile:

$$\text{if } H_0 \text{ is true:} \qquad \hat{p} \;\approx\; N\left(p_0,\ \sqrt{\frac{p_0(1-p_0)}{n}}\right)$$

Notice where every number comes from. The **center** p_0 comes from H_0, not from the data. The standard error also uses p_0, because we are describing the world where H_0 is true. The data enter only once, as the single $\hat{p}$ we observed, which we then place on that curve.

To see how far out on the curve it lands, we turn it into a z-score, exactly as in Weeks 6 and 9 (observation minus mean, divided by standard deviation). This z-score is the **test statistic**:

$$z_{obs} = \frac{\hat{\theta} - \theta_0}{SE} \qquad \text{for a proportion:} \qquad z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

It measures **how many standard errors the estimate is from the value H_0 claims**. If H_0 is true, z_{obs} comes from the standard normal $N(0,1)$, so values near 0 are ordinary and values beyond about ± 2 are rare. The P-value is then an area under the standard normal curve, found with the z-table just as in Week 9.

This is the same machinery as a confidence interval, pointed in a different direction:

	Confidence interval	Significance test
Question	What is the parameter, give or take?	Is the claim $\theta = \theta_0$ believable?
Normal curve centered at	the estimate $\hat{\theta}$	the claimed value θ_0
Uses	$\hat{\theta} \pm z \times SE$	$z_{obs} = (\hat{\theta} - \theta_0)/SE$
Result	a range of believable values for θ	a P-value, then reject or do not reject H_0

Example: has the candidate's support dropped?

It is midterm season (Election Day is Tuesday, November 3). Suppose the polling averages put Candidate X in a Senate race at **52%** support among likely voters. Your campaign suspects X's support is actually lower. You run your own poll of a random sample of $n = 1{,}000$ likely voters, and **490** say they will vote for X: $\hat{p} = 0.490$.

49% is lower than 52%. But Week 8 taught us that two polls of the same state give different answers just because they reach different people. Is a 3-point gap **bigger than ordinary poll-to-poll variation**? That is exactly a significance test.

1. **Assumptions.** A random sample of likely voters, and a large enough sample: $np_0 = 1000(0.52) = 520$ and $n(1-p_0) = 480$, both well above 15.
2. **Hypotheses.** p = the proportion of **all** likely voters who support X. We suspect it is lower than the polls say, so $H_A : p < 0.52$ and $H_0 : p = 0.52$, with $p_0 = 0.52$: **lower-tailed**.

3. **Test statistic.** If H_0 is true, polls of 1,000 voters give values of $\hat{p}$ centered at 0.52 with

$$SE = \sqrt{\frac{0.52 \times 0.48}{1000}} = \sqrt{0.0002496} \approx 0.0158$$

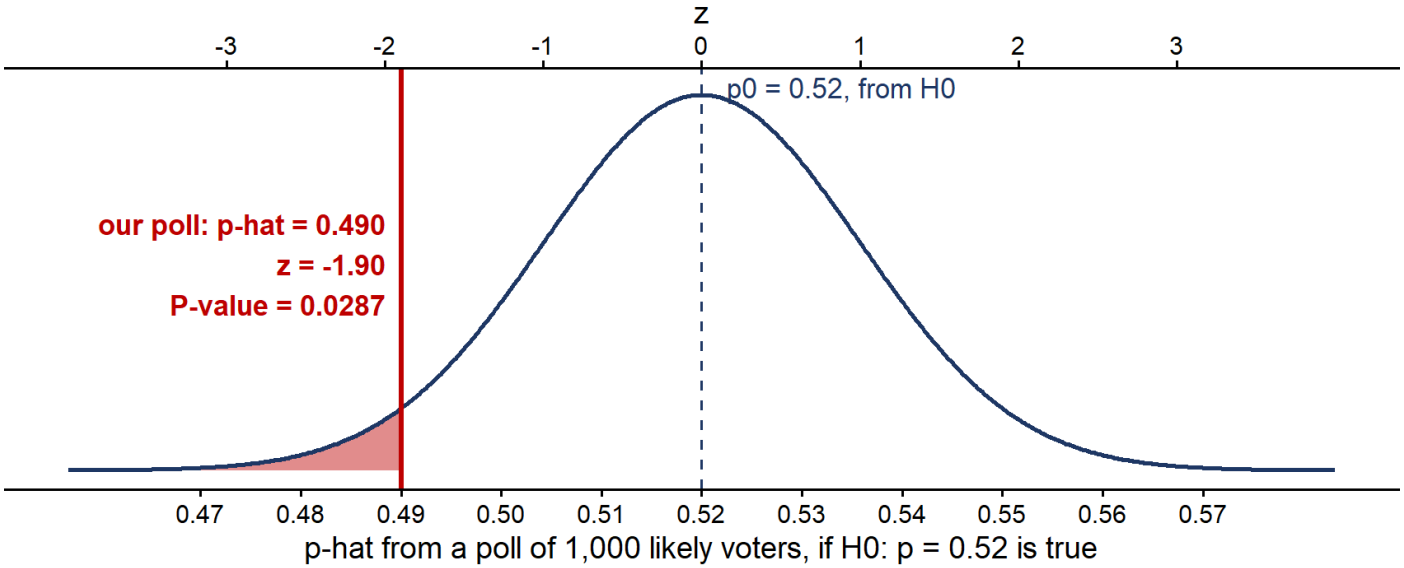
so

$$z_{obs} = \frac{0.490 - 0.52}{0.0158} \approx -1.90$$

Our poll is 1.9 standard errors below what H_0 predicts.

4. **P-value.** $P(Z \leq -1.90 \mid H_0 \text{ true}) = 0.0287$, from the z-table. If X's support really were 52%, only about 3 polls in 100 would come out this low.
5. **Decision.** $0.0287 < 0.05$: **reject** H_0. The poll is evidence that X's support is below 52%.

Here is the same reasoning as a picture. The curve shows where $\hat{p}$ from a poll of 1,000 voters would land **if H_0 were true**; the red line is where ours landed, and the shaded area is the P-value.

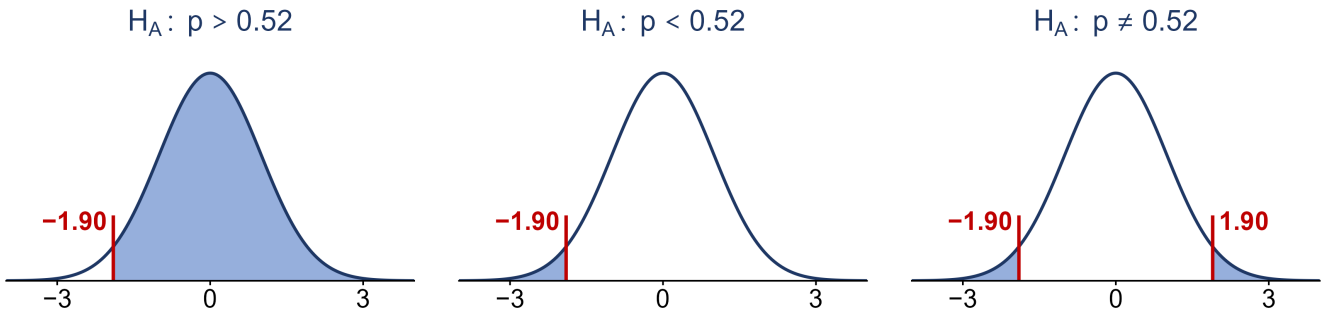


Read the two axes together: the top axis is in the units of the problem (proportions of voters), the bottom axis is the same picture in standard errors. The test statistic is just the poll's position on the bottom axis.

A different claim, the same poll. The headline question on election night is whether X is **losing**, i.e. below 50%. That is a different H_0: $H_0 : p = 0.5$, $H_A : p < 0.5$. Now $SE = \sqrt{0.25/1000} \approx 0.0158$ and $z_{obs} = (0.490 - 0.5)/0.0158 \approx -0.63$, so the P-value is $P(Z \leq -0.63) = 0.2643$. **Do not reject H_0.** The same 49% that is good evidence against "52%" is weak evidence against "50%", because 0.490 sits only 0.63 standard errors below 0.5. A test always answers a question about one specific value θ_0: change θ_0 and you change the question. This is why poll reports say a race is "within the margin of error".

Which tail is the P-value?

"At least as extreme" always means "at least as far in the direction H_A points to". So the shape of H_A decides which area of the curve is the P-value. To see how much it matters, take our poll ($z_{obs} = -1.90$ against $p_0 = 0.52$) and work out the P-value for all three alternatives:



- **Lower-tailed**, $H_A : p < 0.52$: the area **below** z_{obs}, $P(Z \leq -1.90 \mid H_0 \text{ true}) = 0.0287$, read straight from the table.
- **Two-sided**, $H_A : p \neq 0.52$: a poll far out in **either** direction counts as extreme, so the P-value is **both** tails, twice the one-tail area: $2 \times 0.0287 = 0.0574$.
- **Upper-tailed**, $H_A : p > 0.52$: the area **above** z_{obs}, $P(Z \geq -1.90 \mid H_0 \text{ true}) = 1 - 0.0287 = 0.9713$. A poll **below** 52% is no evidence at all that support is **above** 52%, and the huge P-value says so.

At $\alpha = 0.05$, only the lower-tailed test rejects H_0. The two-sided test, with the same data, does not (0.0574): it spreads its doubt over both directions, so it needs a little more evidence. This is why H_A has to come from the question before the poll is run. Choosing the tail after seeing which way the poll went would make any result look more convincing than it is.

Example: do coins land the way they started?

A physics model of coin tossing (Diaconis, Holmes and Montgomery, 2007) predicts that a flipped coin lands on the **same side it started on** a little more than half the time, about 51%, because a real flip wobbles. Bartoš et al. (2025) tested it: 48 people flipped coins 350,757 times, recording the side facing up before and after every flip (and filming it).

Who flipped	Flips
students, for a thesis project	75,036

Who flipped	Flips
coin-flipping marathons (up to 12 hours)	203,440
volunteers recruited online	72,281
total	**350,757**

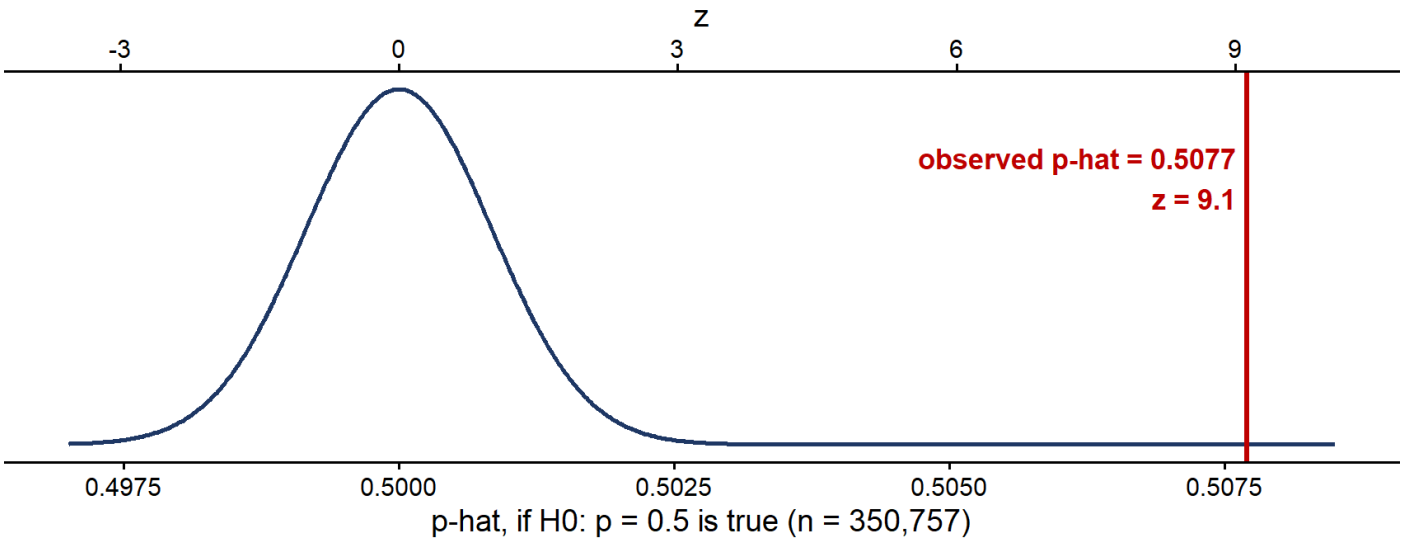
X = 1 if a flip lands on the side it started on, and 0 otherwise. In all, **178,079** of the $n = 350{,}757$ flips did. Test the model's prediction at $\alpha = 0.05$, using the five steps.

1. **Assumptions.** The flips are a random sample of how people flip coins, and n is large: np_0 and $n(1 - p_0)$ are both about 175,000, far more than enough for the normal curve to apply.
2. **Hypotheses.** p = the chance a flip lands on its starting side. The model predicted the direction before any flips were made, so $H_A : p > 0.5$ and $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**.
3. **Test statistic.** $\hat{p} = 178{,}079/350{,}757 \approx 0.5077$ and $SE = \sqrt{0.5 \times 0.5/350{,}757} \approx 0.000844$, so

$$z_{obs} = \frac{0.5077 - 0.5}{0.000844} \approx 9.1$$

   The estimate sits about **9 standard errors** above p_0.
4. **P-value.** $P(Z \geq 9.1 \mid H_0 \text{ true}) \approx 0$. The z-table stops at 3.49, and 9.1 is far beyond it.
5. **Decision.** The P-value is far below 0.05: **reject** H_0. The data are strong evidence that flips land on their starting side more than half the time, just as the model predicted.

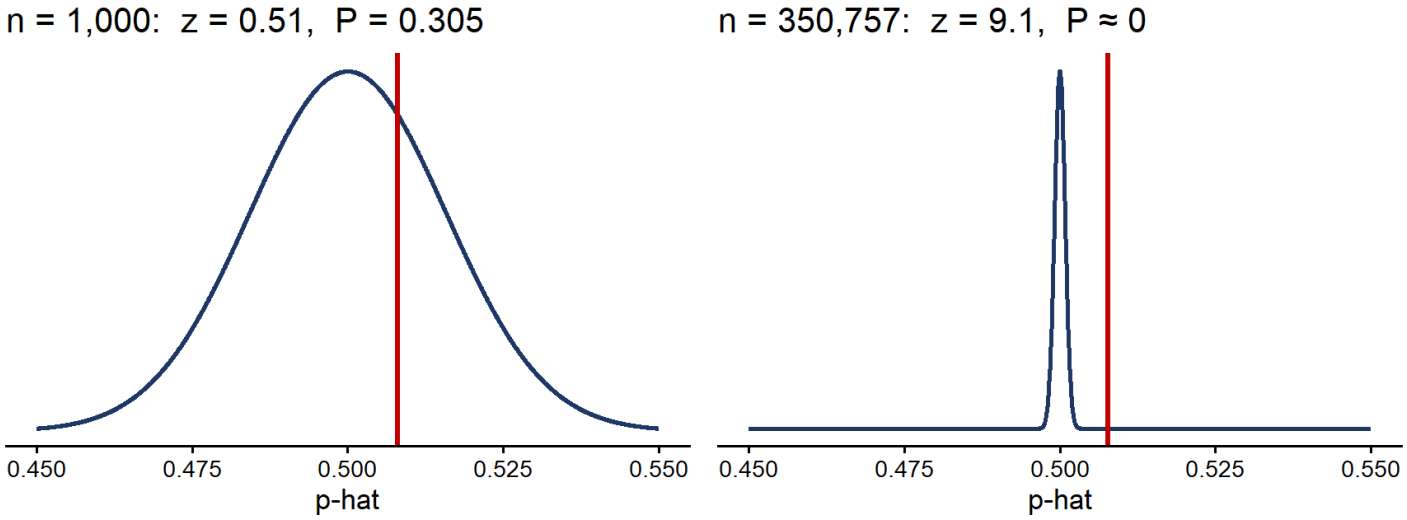
Here is what step 3 looks like on the sampling distribution. The curve is where $\hat{p}$ would land if H_0 were true; the red line is where it actually landed.



Significant is not the same as important

A 50.8% same-side rate instead of 50% is about **1 extra same-side landing in every 130 flips**. Nobody could notice that in a coin toss before a football game. So how can it be "significant"?

Because significance depends on the **sample size**. The standard error has $\sqrt{n}$ on the bottom, so with 350,757 flips the curve of believable $\hat{p}$ values is extremely narrow, and even a tiny difference lands far out in the tail. Suppose instead only 1,000 flips had been made, with the same 50.8% (508 same-side). Then $SE = \sqrt{0.25/1000} \approx 0.0158$, $z_{obs} = (0.508 - 0.5)/0.0158 \approx 0.51$, and the P-value is $1 - 0.6950 = 0.3050$: **not** significant.



Same estimate, opposite conclusions. "Statistically significant" means **"probably not chance"**. It does not mean "big" or "important". Whether a difference matters is a question about the context, and a confidence interval (which shows **how big** the effect is) answers it better than a P-value does.

The five steps of a significance test

Every test in the rest of the course follows the same five steps:

1. **Assumptions.** The data come from a random sample or a randomized experiment, plus any conditions on sample size or on the shape of the population.
2. **Hypotheses.** $H_0 : \theta = \theta_0$ and H_A (upper-tailed, lower-tailed or two-sided), written before looking at the data.

3. **Test statistic.** $z_{obs} = (\hat{\theta} - \theta_0)/SE$, computed as if H_0 were true: how many standard errors the estimate is from θ_0.
4. **P-value.** The probability, if H_0 is true, of a test statistic at least as extreme as z_{obs}, in the tail or tails that H_A points to.
5. **Decision.** If the P-value $< \alpha$, reject H_0 (statistically significant); otherwise do not reject H_0. Then say what it means in the context of the question.

On Wednesday we use these steps to test claims about a population proportion p, and on Friday about a population mean μ.

Try it out

- A friend claims they can predict coin flips. In a test of 10 flips, they call **7** correctly. Write the hypotheses and find the P-value. Is the result significant at $\alpha = 0.05$? *Answer:* $H_A : p > 0.5$, $H_0 : p = 0.5$ (upper-tailed). Under H_0, $X \sim \text{Binom}(10, 0.5)$, so P-value $= P(X \geq 7 \mid H_0 \text{ true}) = (120 + 45 + 10 + 1)/1024 = 176/1024 \approx 0.172$. $0.172 \geq 0.05$: do not reject H_0. Seven of ten is quite common for a guesser.
- Write H_0 and H_A, give θ_0, and name the type of test.
    - A coffee shop says its large cup holds 16 oz on average. Customers suspect it holds less. *Answer:* $H_0 : \mu = 16$, $H_A : \mu < 16$, $\mu_0 = 16$ oz, lower-tailed.
    - A website's sign-up rate used to be 12%. After a redesign, has it changed? *Answer:* $H_0 : p = 0.12$, $H_A : p \neq 0.12$, $p_0 = 0.12$, two-sided.
- A test statistic comes out to $z_{obs} = -2.10$. Find the P-value if H_A is lower-tailed, and if it is two-sided. What is the decision for each at $\alpha = 0.05$? At $\alpha = 0.01$? *Answer:* lower-tailed: $P(Z \leq -2.10) = 0.0179$; two-sided: $2 \times 0.0179 = 0.0358$. Both are below 0.05, so reject H_0 either way at $\alpha = 0.05$. Neither is below 0.01, so do not reject H_0 at $\alpha = 0.01$.

Sources

- Coin flips: Bartoš, F., et al. (2025), "Fair coins tend to land on the same side they started: Evidence from 350,757 flips", *Journal of the American Statistical Association* 120(552), 2118–2127 (preprint on arXiv (https://arxiv.org/abs/2310.04153)). The physics model: Diaconis, P., Holmes, S., and Montgomery, R. (2007), "Dynamical bias in the coin toss", *SIAM Review* 49(2), 211–235.
- The taste test, Candidate X's poll, the chip bag, the coffee shop and the website are made-up examples.

- Textbook: OpenStax *Introductory Statistics 2e*, Chapter 9, Sections 9.1–9.5.